PackSage AI — MoFPI SIH260236

AI-Driven Intelligent Food Packaging Recommendation System

Digital Dossier & Verification Report | ID: 62844b8e-31ea-4186-b186-98384ac1510a

LEGAL & COMPLIANCE NOTICE: Decision-support tool. Requires laboratory testing (ASTM D3985 / ASTM F1249) and shelf-life stability study prior to commercial packaging line commissioning.

1. Commodity & Storage Profile [Evidence-Grade Audit]

Parameter	Input Value	Evidence Badge	Source / Methodology
Commodity	Potato Chips	Verified	MoFPI Commodity Taxonomy
Category	Oily	Verified	Standard Food Classification
Moisture Content	2.0%	User Default	Initial Matrix Moisture
Respiration Rate	N/A mg CO2/kg/h	Estimated	Produce Respiration Model (Q10=2.2)
Storage & Transit	Ambient | road	Real-World	Bharat Supply Chain Simulator
Target Shelf Life	120 Days	User Target	Inverse Shelf-Life Requirement Target

2. Inferred Target Barrier Thresholds (Inverse Solver)

Barrier Parameter	Solved Maximum Permissible Spec	Standard Test Method	Failure Mode Mitigated
Oxygen Transmission (OTR)	≤ 5.0 – 25.0 cc/m²·d	ASTM D3985 (Coulometric)	Lipid Rancidity & Aerobic Spoilage
Water Vapor Transmission (WVTR)	≤ 0.8 – 2.5 g/m²·d	ASTM F1249 (Modulated IR)	Moisture Ingress / Crispness Loss / Caking
Target Thickness	25 - 70 µm	ASTM D6988 / ISO 4593	Pin-hole Puncture & Transit Burst Resistance

3. Recommended Packaging Materials (Ranked by Multi-Criteria Fit)

Rank	Material Structure	Type	Fit Score	OTR	WVTR	Cost Tier	Recyclability
#1	SiOx-coated PET 12µm	high_barrier	0.64	1.0	0.5	4.2/5.0	Recyclable (Class 4)
#2	Met-PET 12µm	metalized	0.64	2.0	0.3	2.5/5.0	Multi-layer
#3	BOPP/Met-PET/LDPE 75µm	metalized	0.63	1.5	0.4	3.2/5.0	Multi-layer
#4	Alu Foil 9µm/LDPE 30µm	foil_laminate	0.62	0.01	0.01	3.5/5.0	Multi-layer
#5	Met-BOPP 20µm	metalized	0.61	5.0	0.5	2.2/5.0	Multi-layer

4. Validation Passport — Laboratory Testing Checklist

Standard Test Protocol	Designation	Target Threshold	Status / Next Step
Oxygen Transmission Rate	ASTM D3985 / ISO 15105-2	Confirm ≤ 5.0 – 25.0 cc/m²·d	Send film swatch to NABL accredited lab
Water Vapor Transmission Rate	ASTM F1249 / ISO 15106-2	Confirm ≤ 0.8 – 2.5 g/m²·d	Verify under 38°C / 90% RH tropical chamber
Heat Seal Integrity & Strength	ASTM F88 / ASTM F2029	≥ 15 N / 15mm seal width	Calibrate pouch sealer dwell & temperature
Puncture & Dart Impact	ASTM D1709 / ASTM F1306	≥ 250 g dart impact	Check truck transit vibration resilience
Accelerated Shelf-Life (ASLT)	MoFPI / IS 10171	120 days equivalent	Run 40°C / 75% RH 30-day incubator trial

5. Supplier Procurement Spec Sheet (RFQ-Ready)

RFQ Field	Supplier Target Specification	Industry Notes
Recommended Material	SiOx-coated PET 12µm	Commercial converter grade
Target OTR Limit	≤ 5.0 – 25.0 cc/m²·d	cc/m²·day @ 23°C, 0% RH
Target WVTR Limit	≤ 0.8 – 2.5 g/m²·d	g/m²·day @ 38°C, 90% RH

Nominal Thickness Range	30 – 55 µm ± 5%	Guaranteed gauge variation < 5%
Sealing Type	Heat Sealable (PE / CPP sealant web)	Seal initiation temp ~ 105-125°C
Food Contact Compliance	FSSAI 2018 / 21 CFR § 177.1520	Migration test certificate mandatory

Verification Passport UUID: 62844b8e-31ea-4186-b186-98384ac1510a
Audit URL: http://localhost:5173/scenario/35f33c60-9600-476a-99a4-fe032c0222e9/results?uuid=62844b8e-31ea-4186-b186-98384ac1510a
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